7.4 Euler's Method

Name _____ Period ____ Date _____

AP Calculus 7.4 · Student sheet

I can…

- use Euler's Method with a given step size.
- organize Euler's Method in a table.
- explain how the step size and concavity affect the approximation.

Vocabulary · fill in the missing word

Euler's Method — $y_{n+1} = y_n + h \cdot f(x_n, y_n)$: follow the _____ line one step at a time

Step size — h, the _____ in x for each step

Part 1 · Follow the arrows

dy/dx = y, starting at (0, 1), steps of 0.5.

x	y	slope = y	next y
0	1		
0.5			
1		—	—

Part 2 · Notes and examples (fill in each note, then work the example)

Euler's Method: $y_{n+1} = y_n + h \cdot$ _____.

Step in x: $x_{n+1} = x_n +$ _____.

Example 1

$\frac{dy}{dx} = x + y$, y(1) = 1. Use h = 0.5 to approximate y(2).

Example 2

$\frac{dy}{dx} = xy + x^2$, y(1) = 2. Use h = 0.25 to approximate y(1.75).

Step size: a smaller h usually gives a _____ approximation.

Example 3

$\frac{dy}{dx} = y$, y(0) = 1. Approximate y(1) with h = 0.5 and with h = 0.25. Compare with e.

Concavity: if the solution is concave up, Euler's Method _____ the solution.

Example 4

Why do both approximations in Example 3 come out too small?

Part 3 · Partner practice

Partner 1

$\frac{dy}{dx} = 2x - y$, $y(1) = 2$. Use $h = 0.1$ to approximate $y(1.2)$.

Partner 2

(AP® Practice) $y = f(x)$ solves $\frac{dy}{dx} = x + 2y$ with $f(0) = 1$. Euler's Method with $h = 0.1$ gives $f(0.3) \approx$

(A) 1.20 (B) 1.76 (C) 2.03 (D) 2.78